

TOPIC 1

DEFINITIONS

Building services refer to the mechanical, electrical, and public health systems that provide water, energy, comfort, safety, and sanitation in buildings.

It involves MEP works

Building services are usually divided into **three major categories**.

1. Mechanical Services

These systems involve **heating, cooling, ventilation, and air movement** in buildings.

Examples include:

- Heating systems
- Air conditioning systems
- Mechanical ventilation
- Elevators and escalators
- Smoke extraction systems

These systems ensure **thermal comfort and air quality** for occupants.

2. Electrical Services

These systems supply and distribute **electrical power and lighting** in a building.

Examples include:

- Electrical power distribution
- Lighting systems
- Backup generators
- Fire alarm systems
- Security systems
- Communication systems (data and internet)

Electrical services ensure buildings have **power, lighting, and communication infrastructure**.

3. Plumbing

These systems deal with **water supply and sanitation**.

Examples include:

- Cold water supply
- Hot water supply
- Plumbing systems
- Drainage systems
- Sewage disposal

- Stormwater drainage
- Fire protection systems

These systems ensure **clean water supply and safe waste removal**.

Guidelines and oversight authorities

In Kenya, the guidelines, standards, and codes for plumbing and sanitation are primarily provided by a combination of national government ministries, regulatory boards, and professional bodies. The key authorities are:

- **Kenya Bureau of Standards (KEBS):** Specifies quality standards, material standards for pipes, fittings, and appliances.
- [Water Services Regulatory Board \(WASREB\)](#): A national regulator that sets national standards, guidelines, and rules for water and sewerage services, ensuring compliance with the Water Act 2016 and the human right to sanitation.
- **Ministry of Water, Sanitation and Irrigation (MoWSI):** Responsible for policy development regarding sanitation management and water supply services.
- **Ministry of Public Works – Mechanical & Drainage Department:** Defines technical specifications, installation methods, and testing standards for plumbing systems.
- **National Construction Authority (NCA):** Regulates the construction industry, ensuring standards in building works, including plumbing.
- **Kenya Water Institute (KEWI):** Develops technical manuals for water supply and sanitation, including the *Practice Manual for Water Supply Services in Kenya*.
- **County Governments:** Responsible for enforcing standards and implementing county-specific water and sanitation strategies and local authority by-laws.
- **Professional Boards (EBK, BORAQS):** Registered Engineers (ERB) and Architects (BORAQS) must sign off on plans, ensuring they meet professional standards.

The Institution of Engineers of Kenya (IEK) +5

Key Regulations and Codes:

- **The National Building Code 2024:** Sets technical requirements for building services, including plumbing.
- **Water Services Regulations, 2021/2024:** Governs the provision of water and sewage services across the country.
- **Public Health Act (Cap 242):** Oversees sanitation and hygiene standards to prevent nuisance.

- **Environmental Management and Coordination (Water Quality)**
Regulations: Ensures safe disposal of wastewater.
- **Sewerage and Sanitation Design Manual:** Used for designing sewerage systems.

CATEGORIES OF WATER CONSUMPTION IN KENYA

Domestic – e.g. drinking, cooking, ablution, sanitation, laundry, garden watering, and bathing
pools.

Trade and industry—e.g. factories, power stations, shops, hotels, hospitals, schools and in offices.

Agriculture -e.g. horticulture, greenhouses, dairies, and livestock farms, etc.

Public use e.g. parks, schools, hospitals, street watering, sewer flushing and firefighting

In truth ‘plumbing’ involves plumbing and drainage system and one needs to differentiate the two:

PLUMBING

Plumbing involves the installation pipes, fixtures, and apparatus to supply potable water to a building.

It encompasses the entire network—including pipes, tanks, and valves—that brings clean water in.

Most common plumbing issues include:

- Leaky pipes: can be caused by pipe corrosion, pipe joint damage, excessive water pressure and a host of other issues.
- Low water pressure: can be a symptom of a more significant problem such as hidden water leaks and blocked pipes.
- Faulty water heater: possible causes include incorrect water pressure, a broken electrical connection and aging equipment.
- Running toilet: a damaged flush valve or faulty flushing handle could be to blame.
- Clogged toilet: often caused by flushing objects that should not be flushed.
- Water heater problems: among the possible causes are sediment build-up and heating element failures.

Main Components of Plumbing

- a) Water Source,
- b) Storage Tanks and distribution ,
- d)Distribution Pipes,

e) Fittings-valves,

f) Fixtures

1. Water Source

There are 3 main sources of water;

- i. Rainfall.
- ii. Surface sources -Lakes, streams, rivers, reservoirs, run off from roofs and paved areas.
- iii. Underground sources- Shallow wells, deep wells, artesian wells, artesian springs, land

Some purification of water is often necessary to remove impurities and possibly to reduce the hardness of water. Purification may involve three major processes as follows;

- a) Sedimentation- Sedimentation is the **process where heavy particles settle at the bottom of a tank due to gravity**. Sludge accumulates at the bottom and clarified water moves to the next stage
- a) Filtration - Filtration is the **process of passing water through porous materials to remove suspended particles**.
- b) Treatment-Chlorination

Filtration of water

i) Pressure filter:

A pressure filter is a closed filtration system where water is forced through filter media under pressure to remove suspended particles and impurities.

Unlike slow sand filters, pressure filters operate inside a sealed tank.

How Pressure Filters Work

Water is **pumped into a pressure vessel** containing filter media such as sand or activated carbon.

The pressure forces water through the media, where:

- Suspended solids are trapped
- Turbidity is reduced
- Some impurities are removed

The filtered water exits through the bottom of the filter.

Filtration Rate

Pressure filters operate at **much higher filtration rates** than slow sand filters.

Typical rate:

5 – 15 m³/m²/hour

Backwashing

Over time, the filter media becomes clogged with particles.

To clean it, **backwashing** is done by:

- Reversing the flow of water
- Flushing out trapped particles

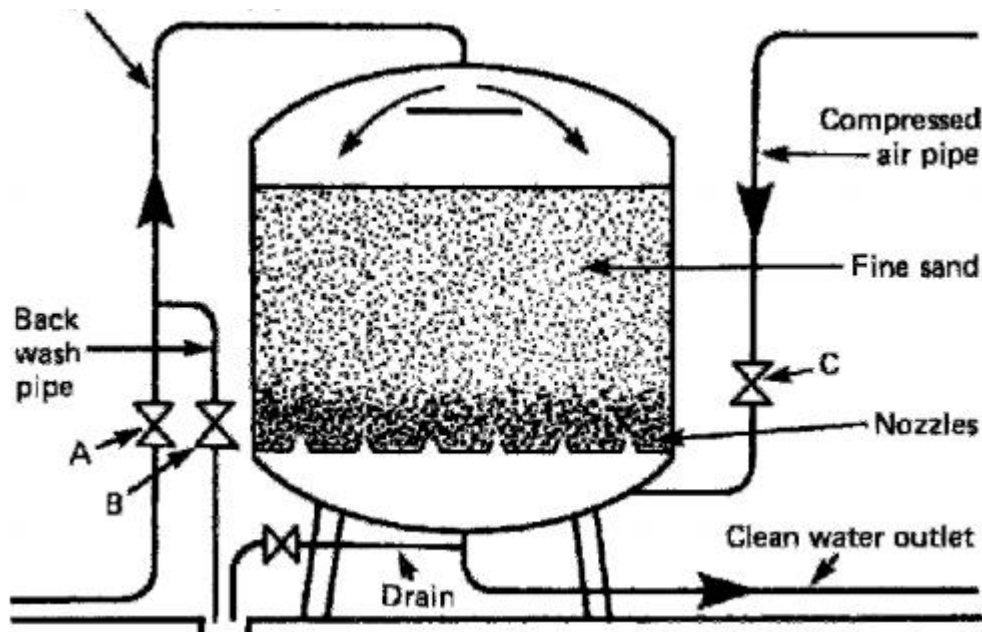
Advantages

- Requires less space
- Higher filtration rate
- Compact design
- Suitable for buildings and industries

Disadvantages

- Requires pumps and energy
- Higher operational cost
- Requires regular maintenance

To backwash, valve A is closed and valves B and C opened.



c) Slow sand filter bed:

A **slow sand filter** is a water filtration system where water passes slowly through a bed of sand to remove impurities and microorganisms. It is one of the oldest and most effective natural water purification methods.

How a Slow Sand Filter Works

Water moves **slowly by gravity** through layers of sand and gravel. During this process:

1. Suspended particles are **physically trapped** in the sand.
2. Microorganisms are **removed by biological action**.

Main Components of a Slow Sand Filter

A typical slow sand filter consists of:

1. **Supernatant water layer**
 - Raw water stored above the sand bed.
2. **Sand bed**
 - Fine sand that performs most of the filtration.
3. **Gravel layer**
 - Supports the sand and allows water to flow evenly.
4. **Underdrain system**
 - Collects filtered water.

Filtration Rate

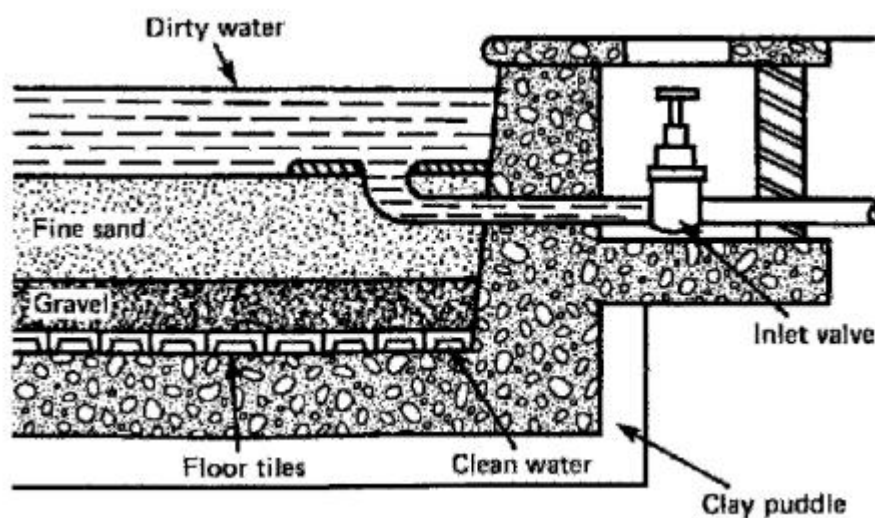
Slow sand filters operate at a **very low filtration rate**:

0.1 – 0.3 m³/m²/hour

Because of the slow rate, they require **large surface areas**.

Filter beds can

occupy large areas and the top layer of sand will require removal and cleaning at periodic intervals.

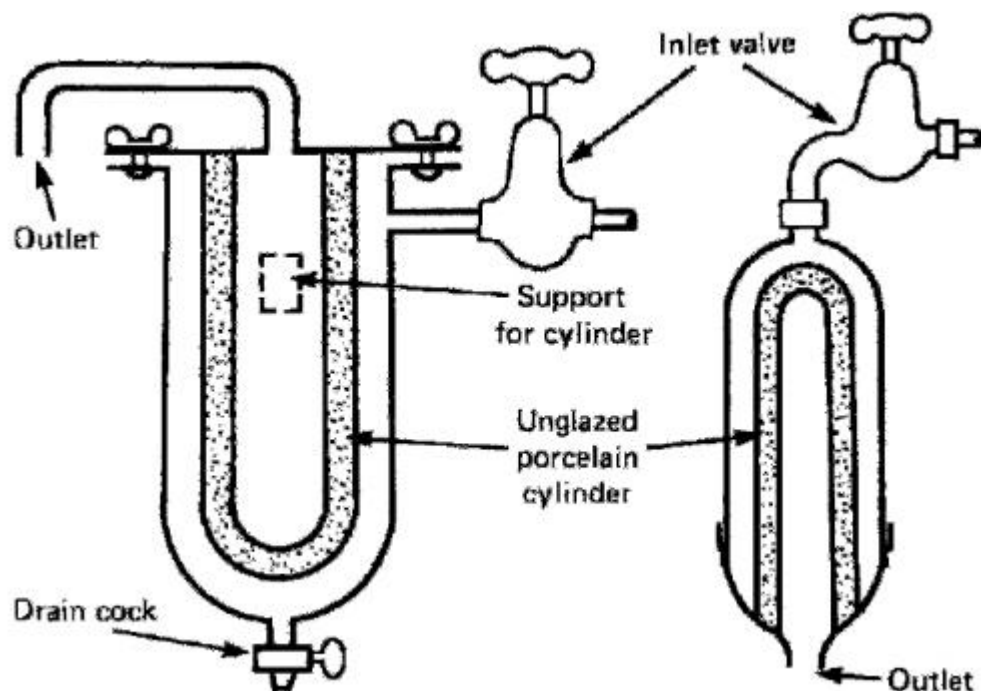


Common Applications

Slow sand filters are often used in:

- Rural water treatment plants
- Small communities
- Developing regions
- Small municipal systems

c) **Small domestic filter:** the unglazed porcelain cylinder will arrest very fine particles of dirt and even micro-organisms. The cylinder can be removed and sterilized in boiling water for 10 minutes



Sterilization and softening of water.

a) **Sterilization of Water**

Definition

Sterilization (also called **disinfection**) is the **process of destroying harmful microorganisms in water to make it safe for drinking.**

Untreated water may contain microorganisms such as:

- Bacteria
- Viruses
- Parasites

These microorganisms can cause **waterborne diseases** such as cholera, typhoid, and dysentery.

a) **Sterilization by Chlorine Injection**

Chlorination involves **adding small quantities of chlorine to water to destroy harmful microorganisms and organic matter.**

Chlorine is widely used because it is:

- Highly effective against bacteria
- Relatively inexpensive
- Easy to apply
- Provides a **residual disinfectant effect**

Process of Chlorination

1. Water is first treated through **filtration** to remove suspended particles.
2. A measured amount of **chlorine is injected into the water.**
3. Chlorine reacts with microorganisms and **destroys their cellular structure.**
4. The treated water is stored for a short **contact period** to allow disinfection to occur.

Chlorine Dosage

The typical chlorine dosage for drinking water is:

0.1 – 0.3 parts per million (ppm)

This amount is sufficient to:

- Kill harmful bacteria
- Maintain a small **residual chlorine level** in the distribution system

Residual chlorine helps **prevent recontamination of water during storage and distribution.**

Advantages of Chlorination

- Effective at killing pathogens
- Low cost
- Provides residual protection
- Suitable for large-scale water supply systems

Limitations

- Excess chlorine can produce **unpleasant taste and odor**
- High chlorine levels may form **disinfection by-products**
- Requires proper dosage control

b) **Softening of Hard Water**

Softening of hard water by base exchange process: sodium zeolites exchange their sodium base for calcium (chalk) or magnesium bases in the water. Sodium zeolite plus calcium

carbonate or sulphate becomes calcium zeolite plus sodium carbonate or sulphate. To regenerate, salt is added; calcium zeolite plus sodium chloride (salt) becomes sodium zeolite plus calcium chloride which is flushed away

Definition

Water softening is the **process of removing dissolved minerals such as calcium and magnesium that cause water hardness.**

Hard water commonly contains:

- Calcium carbonate
- Calcium sulphate
- Magnesium salts

Problems Caused by Hard Water

Hard water can cause several problems in buildings, including:

- Formation of **scale deposits in pipes and boilers**
- Reduced efficiency of water heaters
- Increased soap consumption
- Stains on plumbing fixtures
- Reduced lifespan of appliances

Softening by Base Exchange Process (Zeolite Process)

Principle

The **base exchange process** uses a material called **sodium zeolite**, which has the ability to **exchange sodium ions for calcium and magnesium ions present in hard water.**

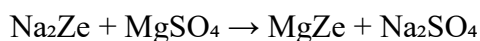
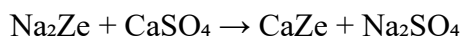
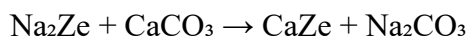
This process removes hardness from water.

Chemical Reaction

When hard water passes through sodium zeolite it exchanges its sodium ions with Ca^{+} Mg^{+} :

Sodium zeolite + calcium salts $\rightarrow$ calcium zeolite + sodium salts

Example reactions:



In this reaction:

- Calcium and magnesium ions are **removed from water**
- Sodium ions replace them

As a result, the water becomes **softened.**

Regeneration of Zeolite

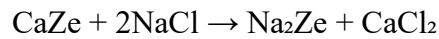
Over time, the zeolite becomes **exhausted** because it becomes saturated with calcium and magnesium ions.

To restore its effectiveness, the zeolite must be **regenerated**.

Regeneration Process

Salt (sodium chloride also known as brine) is added to the zeolite bed.

Chemical reaction:



During this process:

- Sodium ions from salt replace calcium ions in the zeolite
- Calcium chloride is formed
- The calcium salts are then **flushed away during backwashing**

The zeolite is now ready to **soften water again**.

Advantages of the Base Exchange Process

- Highly effective in removing water hardness
- Simple and reliable method
- Suitable for domestic and industrial use
- Can be regenerated and reused many times

Limitations

- Requires regular regeneration using salt
- Does not remove bacteria or other impurities
- May increase sodium content in water

Sterilization (Disinfection)

- Purpose: Kill harmful microorganisms
- Method: Chlorine injection
- Typical dosage: **0.1 – 0.3 ppm**

Softening

- Purpose: Remove hardness (calcium and magnesium)
- Method: **Base exchange process using sodium zeolite**
- Regeneration: Using **salt (sodium chloride)**

2. Storage and distribution of water

In building services engineering, water must be **stored and distributed properly** to ensure a reliable supply to all parts of a building.

Water storage and distribution systems are designed to:

- Provide **continuous water supply**

- Maintain **adequate pressure**
- Meet **peak water demand**
- Provide **water during supply interruptions**
- Ensure **water is available for fire protection**

In many areas, water supply from municipal sources may be **intermittent**, making storage facilities essential.

Water storage refers to the **collection and holding of water in tanks or reservoirs before it is distributed for use in a building**.

Storage ensures that water is available even when the **main supply is interrupted or demand increases**.

Types of Water Storage Tanks

Water in buildings is usually stored in two main types of tanks.

i) Underground Storage Tanks

These tanks are located **below ground level**. The purpose of which is to store water received from municipal supply or boreholes and act as the **main water reservoir** for the building

Characteristics

- Usually made of **reinforced concrete, steel, or plastic**
- Large storage capacity
- Protected from contamination and temperature changes

Advantages

- Large storage volume
- Protected from sunlight and temperature changes
- Does not occupy useful building space

ii) Overhead Storage Tanks

These tanks are installed **above the building, usually on the roof or a tower**. The purpose of which is to provide **gravity-based water distribution** and maintain water pressure in the building

Characteristics

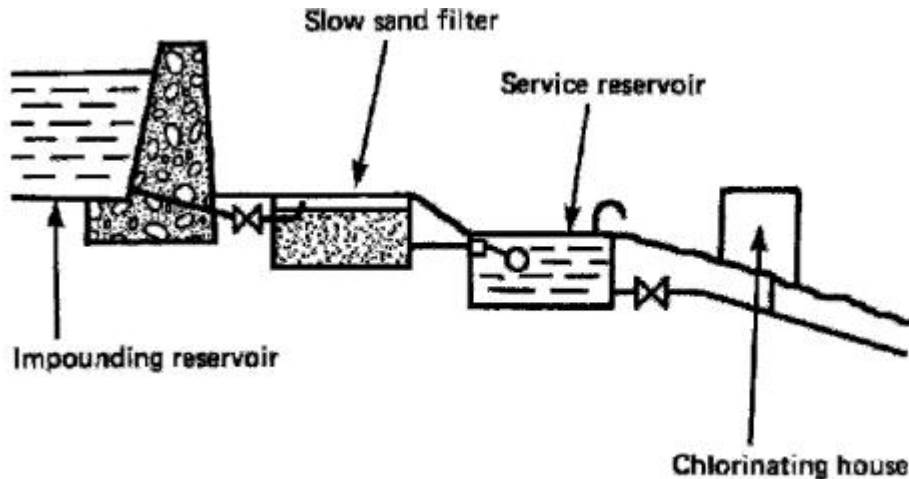
- Usually made from **plastic (PVC), fiberglass, or steel**
- Located at the **highest point of the building**

Advantages

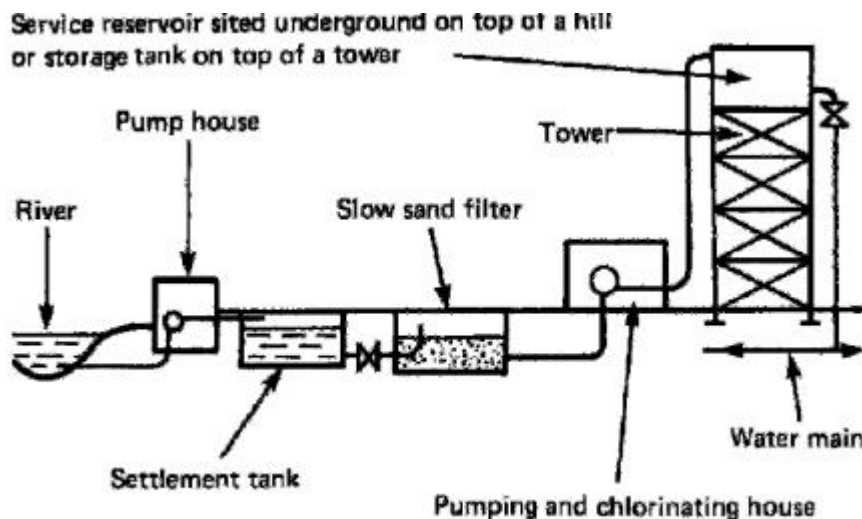
- Provides water pressure through gravity
- Ensures continuous supply during pump failure

Water distribution refers to the **network of pipes that carry water from the storage tank or main supply to various fixtures in a building.**

a) **Gravitational distribution:** the water from upland gathering grounds is impounded in a reservoir. From this point the water is filtered and chlorinated before serving an inhabited area at lower level. There are no pumping costs



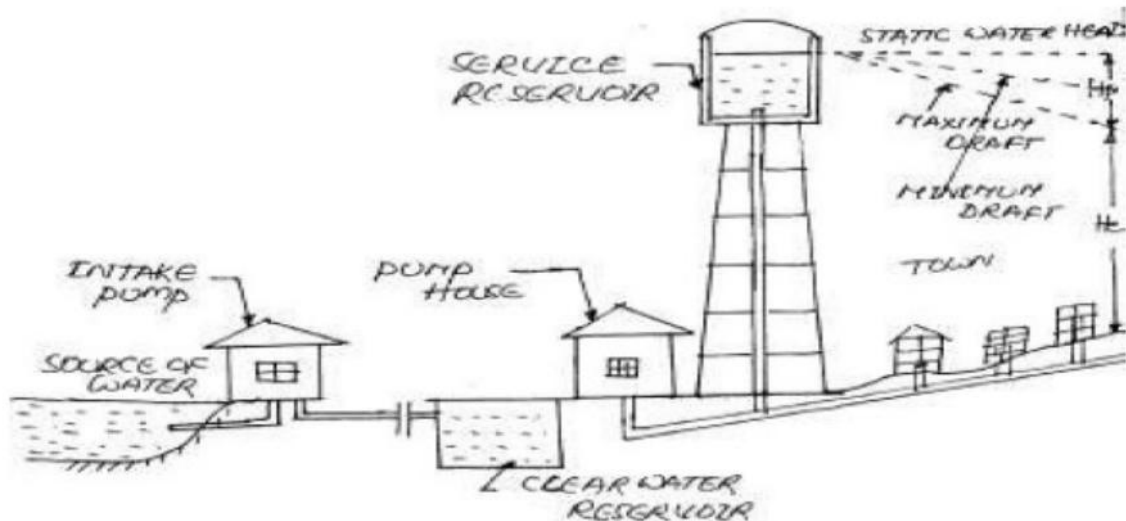
b) **Pumped distribution:** water extracted from a river is pumped into a settlement tank, subsequently filtered and chlorinated. Pump maintenance and running costs make this process more expensive than gravity systems.



c) **Combined gravity and pumping system:**

- Most common system.
- Treated water is pumped and stored in an elevated distribution reservoir.
- Then supplies to consumer by action of gravity.
- The excess water during low demand periods gets stored in reservoir and get

- supplied during high demand period.
- Economical, efficient and reliable system



SUPPLY OF WATER TO THE BUILDING.

Water is supplied by local authority to estates via a water main. A water main is a water supply pipe usually serving water to an estate. It's installed by the Local Authority. The water mains are usually dig and laid underground. UPVC pipes are pigmented blue for easy identification in future excavations and cast iron has a blue plastic tape attached for the same reason.

Connection to Water Main, Service and Supply Pipe.

Connection to water main is done by the local authority upon written notice for connection to their supply main. *The main is drilled and tapped live with special equipment, which leaves a plug valve ready for connection to the communication pipe.*

A goose neck or sweeping bend is formed at the connection to relieve stresses on the pipe and valve. The pipe that connects the water mains and brings water to or close to the property boundary is called **a service pipe or communication pipe.**

At or close to the property boundary, a stop valve is located with an access compartment and cover at ground level. A meter may also be located at this point.

The purpose of the stop valve is to enable the local authority to disconnect the water supply in case of non-payment or maintenance.

The pipe that runs water from the stop valve into the building is called the **supply pipe.** (*A supply pipe is any pipe, maintained by the consumer, which is subject to water pressure from the authority's mains. The supply pipe runs from the stop valve to, and into, the building;*

maintenance is the responsibility of the land/building owner.) At the point the supply pipe enters the building, there should be a stop valve as well. This is to allow the user to disconnect in case they want to undertake repairs and maintenance. The service/communication and supply pipe should be snaked to allow for settlement in the ground.

Distributing pipe is any pipe (other than an overflow or flush pipe) that conveys water from a storage cistern or from hot water apparatus supplied from a feed cistern and under pressure from that cistern

Water Mains

In Kenya, water supplied to buildings generally comes from **county water service providers** (water utilities) that operate under the **national water sector framework**.

Examples of such utilities include: Nairobi City Water and Sewerage Company, Mombasa Water Supply and Sanitation Company, Kisumu Water and Sanitation Company

These companies distribute treated water through **public water mains**, which supply estates, institutions, and individual buildings.

The building water supply system therefore consists of **two main sections**:

1. **Public supply system** – managed by the water service provider
2. **Private plumbing system** – maintained by the property owner

A **water main** is a **large underground pipe installed by the local water service provider to distribute treated water to different estates or neighborhoods**.

Water mains transport water from:

- Water treatment plants
- Pumping stations
- Storage reservoirs

to residential, commercial, and industrial areas.

Connection to the Water Main

Connection of a building to the water main must be done **by the water service provider**.

The property owner normally applies for a **new water connection**.

Typical Procedure in Kenya

1. The property owner submits an application to the water company.
2. The utility inspects the property and approves the connection.
3. The main pipe is **drilled and tapped while still under pressure** using special equipment.
4. A **ferrule or tapping saddle** is installed to create the connection.

5. A **service pipe** is connected to carry water toward the property.

This process ensures that the **main supply to other customers is not interrupted**.

Goose Neck (Swan Neck)

At the connection point to the water main, a **goose neck** or **swan neck** is installed.

Definition

A **goose neck** is a **short curved section of pipe** used to connect the water main to the **service pipe**.

Purpose

The goose neck:

- Provides **flexibility**
- Absorbs **movement due to soil settlement**
- Prevents **damage to the pipe joint**

This is particularly important in Kenya where **road excavation, soil movement, and heavy traffic loads** may occur.

i) Service Pipe (Communication Pipe)

The **service pipe**, also known as the **communication pipe**, is the pipe that **connects the water main to the property boundary**.

It is usually installed and maintained by the **local authority**.

Functions of the Service Pipe

The service pipe:

- Carries water from the **public water main**
- Supplies water **up to the property boundary**
- Connects the public supply to the private plumbing system

Stop Valve and Water Meter

At or near the **property boundary**, the service pipe is fitted with:

- A **stop valve**
- A **water meter** (in many cases)

These are installed inside a **meter box or access chamber** located at ground level.

Stop Valve

A **stop valve** is a control valve used to **shut off the water supply**.

Purpose

The stop valve allows:

- The local authority to **disconnect supply in case of non-payment**
- Water supply to be **stopped during maintenance or repairs**

- Control of water flow into the property

Water Meter

A **water meter** may also be installed near the property boundary to:

- Measure the **amount of water consumed**
- Enable **billing based on usage**

ii) Supply Pipe

A **supply pipe** is the pipe that **carries water from the stop valve at the property boundary into the building**.

Unlike the service pipe, the supply pipe is:

- **Installed and maintained by the property owner**
- Considered part of the **private plumbing system**
- **Internal Stop Valve**

Where the supply pipe enters the building, another **stop valve** should be installed.

This allows the building owner or occupants to:

- Shut off the water supply
- Carry out **repairs or maintenance**
- Prevent water damage during pipe failures

Pipe Flexibility and Settlement

Both the **service pipe** and **supply pipe** should be laid with slight curves or “snaking”.

This allows the pipe to:

- Adjust to **ground settlement**
- Prevent **pipe breakage**
- Reduce **stress on pipe joints**

Straight rigid pipes are more likely to **crack when soil movement occurs**.

iii) Distributing Pipes

A **distributing pipe** is any pipe that **conveys water plumbing fixtures within the building**.

These pipes form part of the **internal plumbing distribution network**.

Function of Distributing Pipes

Distributing pipes deliver water to:

- Wash basins
- Showers
- Sinks
- Toilets
- Water heaters

- Other plumbing fixtures

They carry water **under pressure from storage tanks or feed cisterns.**

Typical Water Supply Path to a Building

The flow of water to a building generally follows this sequence:

Water Treatment Plant

↓

Water Main (Public Supply)

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Service Pipe / Communication Pipe

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Stop Valve and Water Meter

↓

Supply Pipe

↓

Internal Stop Valve

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Storage Tank or Distribution Pipes

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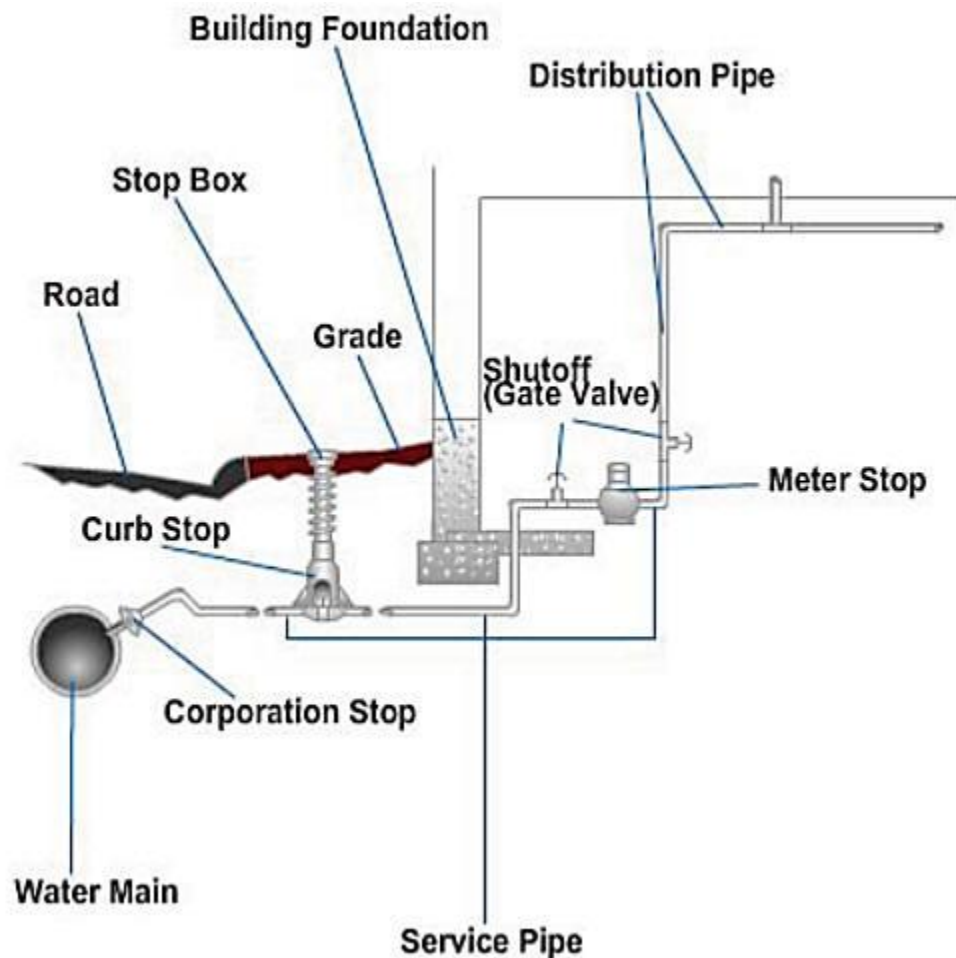
Plumbing Fixtures

Summary

The water supply system to a building consists of several components working together:

- **Water Main** – public supply pipe serving an estate
- **Service (Communication) Pipe** – connects the main to the property boundary
- **Stop Valve and Meter** – controls and measures water supply
- **Supply Pipe** – carries water into the building
- **Distributing Pipes** – supply water to internal fixtures

Proper installation and maintenance of these components ensures **safe, reliable, and efficient water supply to buildings.**



SUPPLY OF WATER IN THE BUILDING.

Broadly divided into:

- Cold Water Supply
- Hot Water Supply.

Cold water supply is nothing but an external water supply. However, cold water supply system can also use filter, water softener appliances, or any other fixture. The connection for the cold-water system is done in such a way that other appliances could receive it through fixtures and taps. Such appliances include sinks, hot water heaters, faucets, bathtubs, showers etc.

Hot water system; There are certain appliances that can be used to provide hot water, such as water heaters. Cold water supply system supplies a volume of water to such appliances, where they heat the water and provide hot water

1. Cold Water Supply.

Broadly divided into:

- Direct system of cold-water supply

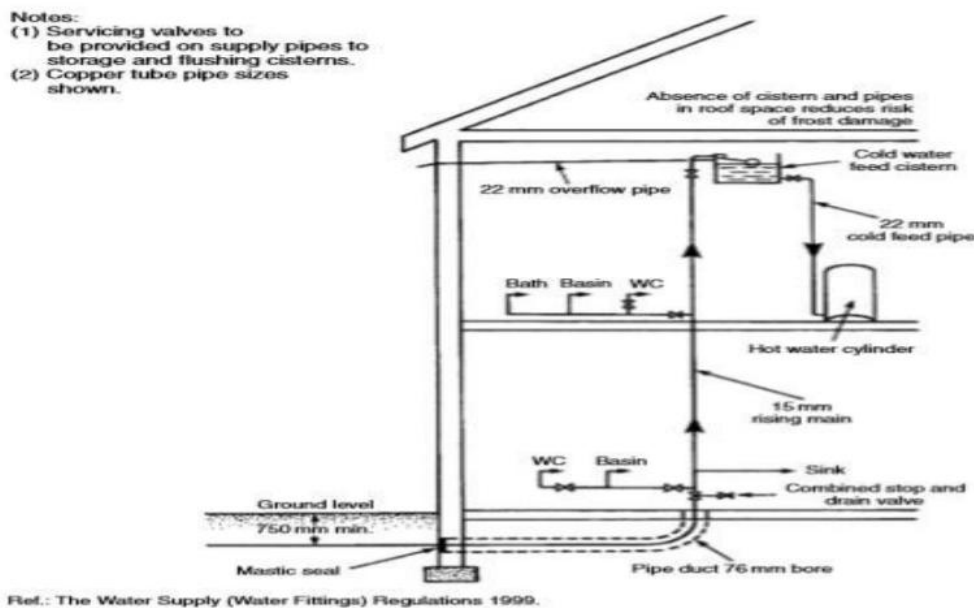
- Indirect system of cold-water supply.

A. Direct System of Cold-Water Supply

In this system, water from the rising main is tapped to have drinking water at every draw-off point. There is no storage for other uses except a storage cistern supplying the hot water cylinder whose capacity is not more than 115 litres capacity. Drinking/portable water is therefore available at every draw-off point and maintenance valves should be fitted to isolate each section of pipework. With every outlet supplied from the main, the possibility of back siphonage must be considered. (**Back siphonage is the reverse flow of water into water supply cause of negative pressure in supply line**)

Back siphonage can occur when there is a high demand on the main. Negative pressure can then draw water back into the main from a submerged inlet, e.g., a rubber tube attached to a tap or a shower fitting without a check valve facility left lying in dirty bath water.

Below is a diagram illustrating direct system of cold-water supply



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Advantages

- Small cold water storage cistern to feed hot water tanks required. This reduces builders work to support larger storage tanks.
- A uniformly high-pressure supply to all hot and cold-water outlets some distance below the pumped head of the supply
- Drinking water is available at all points
- Reduces pipework.

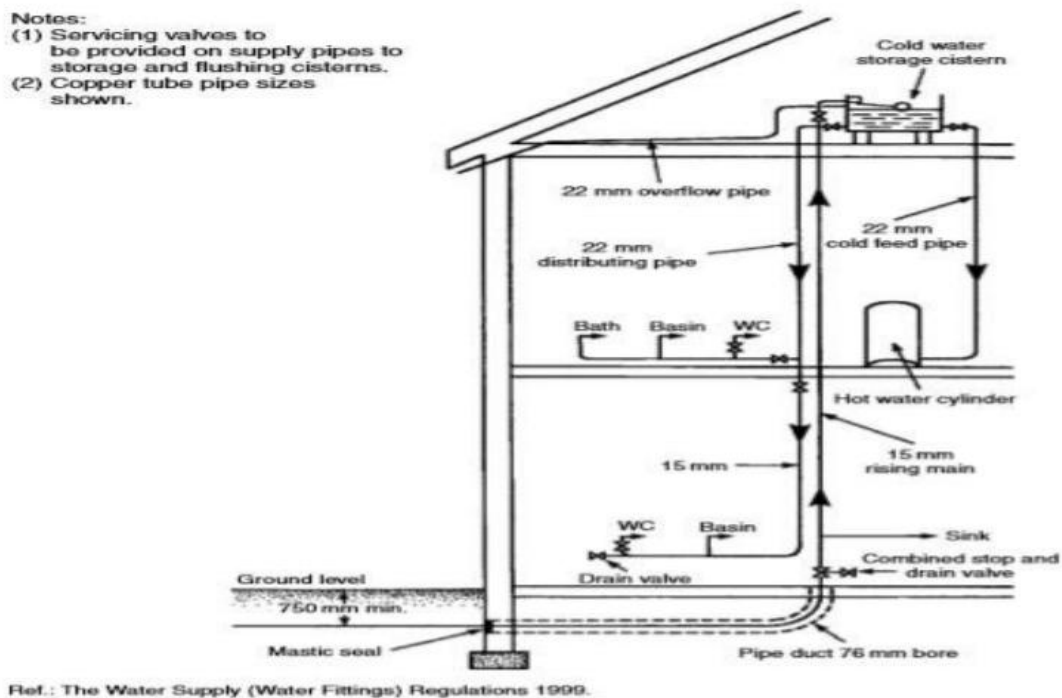
Disadvantages

- In case of pipe bursts(mains) there is water shortage
- There is risk of water contamination due to back siphonage.
- High pressure from the mains provides higher level of noise.
- High pressure from the mains contributes to wear on fittings which necessitates comparatively frequent inspection, maintenance and repair of many valves and controls to the system

B. Indirect cold-water supply

In this system, the rising main service pipe serves only one draw-off point often the kitchen sink. All other cold-water draw-off points are supplied from the storage cistern.

The cold-water storage cistern has a minimum capacity of 230 litres, for location in the roof space. In addition to its normal supply function, it provides an adequate emergency storage in the event of water main failure. The system requires more pipework than the direct system and is therefore more expensive to install, but uniform pressure occurs at all cistern-supplied outlets. The water authorities prefer this system as it imposes less demand on the main. Also, with fewer fittings attached to the main, there is less chance of back siphonage. Other advantages of lower pressure include less noise and wear on fittings, and the opportunity to install a balanced pressure shower from the cistern.



Advantages of indirect supply

- Reserves supply in case of mains failure
- With fewer fittings attached to the main, there is less chance of back siphonage therefore reducing risk of contamination.
- The internal distribution pipes accrue lower pressure which translate less noise and wear on fittings, and the opportunity to install a balanced pressure shower from the cistern.
- The air gap between the supply pipe and the water level in the cistern acts as an effective barrier to backflow into the mains supply, which can cause contamination

Disadvantages

- Requires more pipe work
- The considerable weight of a filled cistern has to be supported at high level
- The inconvenience of access to the cistern for inspection and maintenance
- The possibility of the cistern overflowing
- The need to insulate effectively the cistern and its associated pipework against freezing
- Animals and insects may enter the cistern and contaminate the water held in the open Cistern

C. Hydro-pneumatic cold water supply system

Hydro-pneumatic system is a variation of direct pumping system. A **hydro-pneumatic cold water supply system** is a water supply system where pumps and compressed air in a pressure tank are used to maintain constant water pressure and distribute cold water throughout a building.

Hydro-pneumatic system generally eliminates the need for an overhead tank and supply water at a much higher pressure than available from overhead tanks particularly on the upper floors, resulting in even distribution of water at all floors

The system uses **three main elements**:

1. **Water pump**
2. **Pressure tank (hydro-pneumatic tank)**
3. **Compressed air cushion**

Step-by-step operation

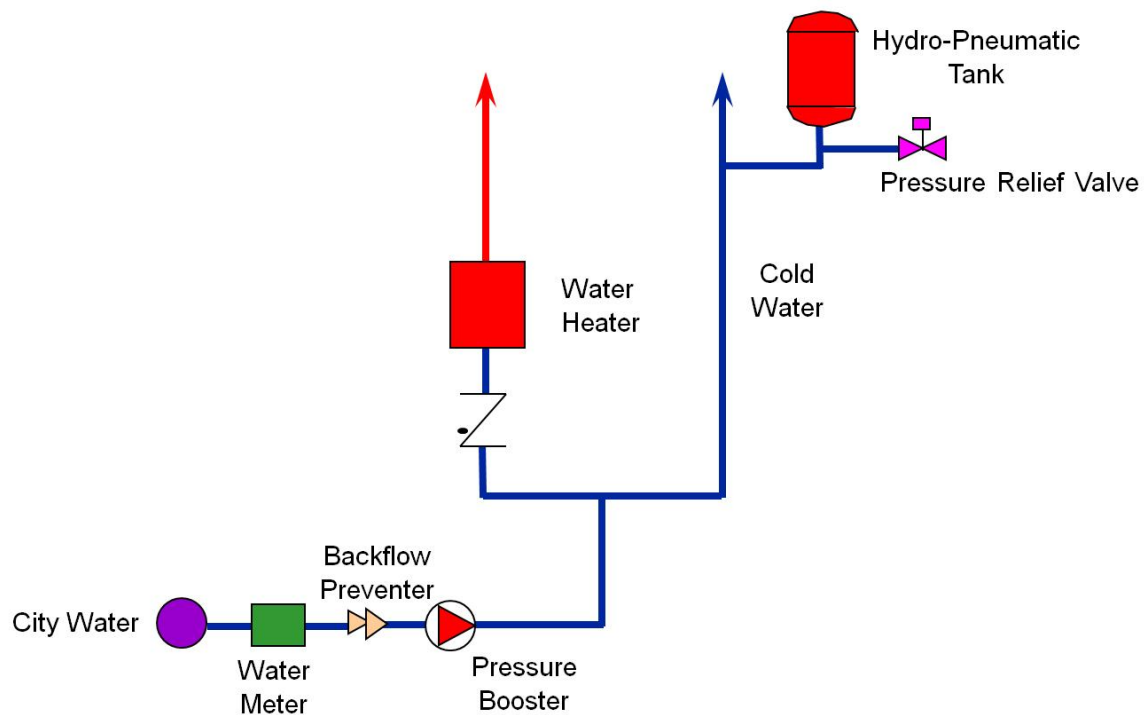
- i. Water from the **main supply or storage tank** enters the system.
- ii. A **pump pushes the water into a sealed pressure tank.**

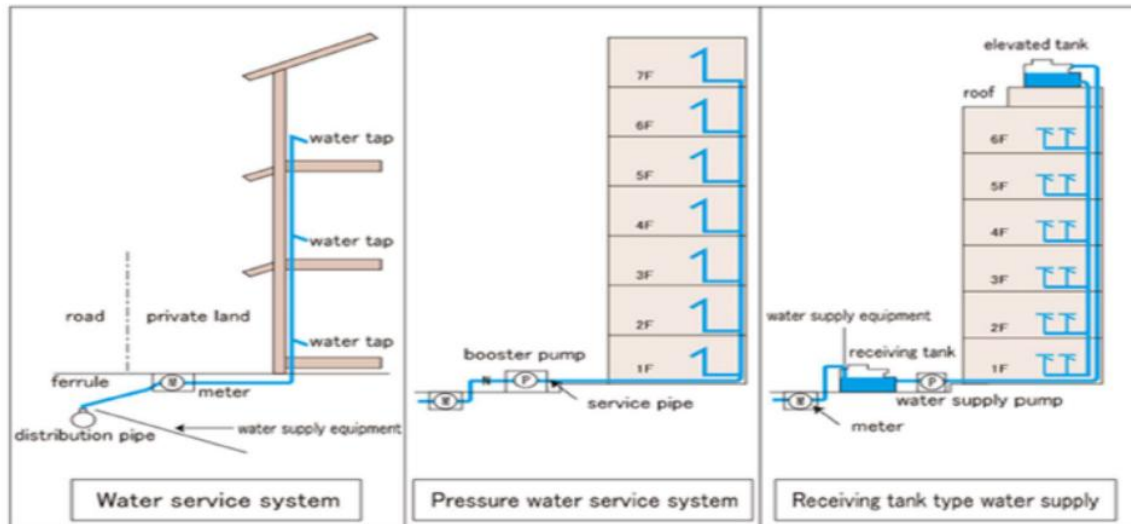
- iii. Inside the tank there is **compressed air above the water**.
- iv. As water enters, the **air compresses and stores pressure energy**.
- v. When taps are opened, the **compressed air pushes water out through the distribution pipes**.
- vi. When pressure drops below a set level, the **pump automatically starts again**.

Basically, the system is configured into 5 types they are:

- 1) Single booster system
- 2) Zone- divided system
- 3) Roof tanks system
- 4) Series- connected systems with intermediate break tanks
- 5) Series- connected system

Hydropneumatic Tank Installation

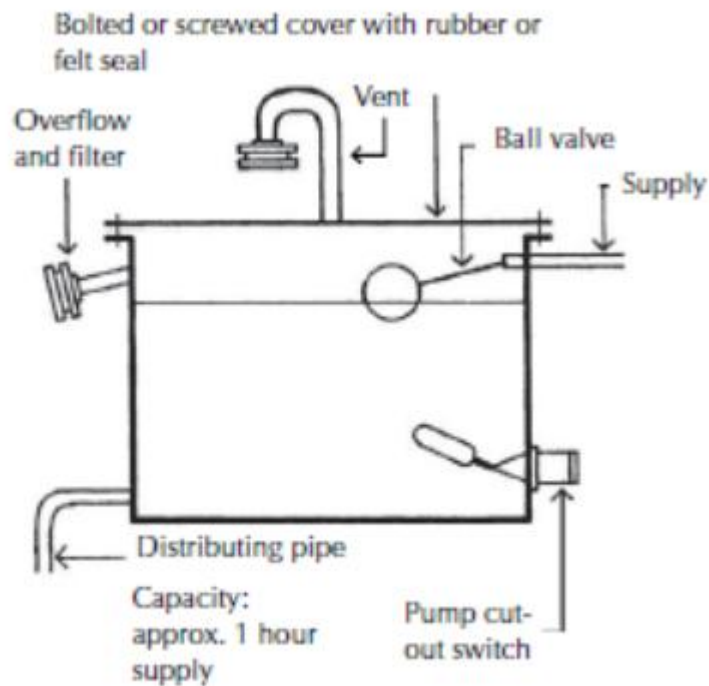




Cold Water Storage Tanks

Tanks can be manufactured from galvanized mild steel (large nondomestic capacities), polypropylene or glass reinforced plastics. They must be well insulated and supported on adequate bearers to spread the concentrated load. Plastic tanks will require uniform support on boarding over bearers. A dustproof cover is essential to prevent contamination.

For large buildings, cisterns are accommodated in a purpose- made plant room at roof level or within the roof structure. This room must be well insulated and ventilated, and be provided with thermostatic control of a heating facility. Where storage demand exceeds 4500 litres, tanks must be duplicated and interconnected. In the interests of load distribution this should be provided at much lower capacities. For maintenance and repairs each cistern must be capable of isolation and independent operation.



Drinking water storage cistern.

2. Hot Water Supply Systems

Hot Water Supply Systems in Kenya

The selection of heating energy sources depends on **energy availability, building scale, operational cost, and environmental considerations.**

Hot water in buildings is heated using different energy sources depending on **availability, cost, and building type.**

i) Electric Water Heating

This is the most common system in urban residential buildings.

How it works

Electric heating elements inside a tank heat water.

Types: Instant electric heaters and storage electric heaters

Advantages: Easy to install, reliable, widely available

Disadvantages: high electricity cost, requires constant power supply

A **hot water supply system** is a system designed to **generate, store, control, and distribute** heated water to various building fixtures while maintaining required temperature, pressure and flow conditions.

The design of hot water systems is an important component of **building services engineering**, particularly in facilities where hygiene, comfort, and operational processes require controlled thermal water supply.

Typical buildings requiring hot water include:

- Residential apartments
- Hotels and hospitality facilities
- Hospitals and healthcare institutions
- Educational institutions
- Industrial facilities
- Commercial kitchens and laundries

From an engineering perspective, the design of hot water systems must consider:

- **Thermodynamic efficiency**
- **Hydraulic performance**

Energy sources include

1. Electric Water Heating

Electric heating systems operate through **electrical resistance heating elements** that convert electrical energy into thermal energy.

Electric heaters are commonly classified as:

a) Storage Water Heaters

Water is heated and stored in insulated tanks. Typical tank sizes: 50 L, 100 L, 200 L, 300 L
These systems rely on **thermostatic control** to maintain desired temperatures.

b) Instantaneous (Tankless) Electric Heaters

Water is heated **on demand** using high-power heating elements.

Advantages:

- Eliminates standby heat losses
- Compact design

Limitations:

- Requires high electrical power
- Limited simultaneous demand capacity

In Kenya, electricity costs often make **electric heaters expensive for large-scale hot water demand**.

2. Solar Water Heating Systems

Solar water heating systems convert **solar radiation into thermal energy** using collectors.

Kenya lies near the **equator**, receiving average solar radiation of approximately **4–6 kWh/m²/day**, making solar heating highly viable.

Large buildings in Kenya are influenced by policies encouraging solar water heating to reduce electricity consumption.

Components of Solar Water Heating Systems

1. Solar collectors
2. Storage tanks
3. Circulation pumps (in active systems)
4. Heat exchangers
5. Backup heating systems

Solar System Configurations

i. Thermosiphon Systems

Operate through **natural convection circulation**.

Hot water rises while cold water sinks.

Advantages:

- No pumps required
- Simple operation

Limitations:

- Requires elevated storage tanks

ii) Forced Circulation Systems

Use pumps to circulate water between collectors and storage tanks.

Suitable for:

- Large buildings
- Multi-storey facilities

3. Gas Water Heating

Not so common in Kenya. Gas water heaters utilize **combustion of LPG or natural gas**.

Gas heating provides:

- Rapid heating
- High thermal output
- Lower operational costs compared to electric heating

However, systems require:

- Proper ventilation
- Gas leak detection
- Safety compliance

4. Heat Pump Water Heating

Heat pumps extract heat from **surrounding air to heat water**.

Heat pumps operate based on **refrigeration cycle thermodynamics**, transferring heat from ambient air to water.

The cycle includes:

1. Evaporation
2. Compression
3. Condensation
4. Expansion

Advantages

- Very energy efficient
- Lower energy consumption

Disadvantages

- High installation cost
- Requires technical maintenance

Classification of Hot Water Supply Systems

Hot water systems may be classified according to **location of heating equipment and distribution method**.

a) Centralized Hot Water Systems

A centralized system produces hot water in **one central plant room** and distributes it through the building.

Components include:

- Central boilers or heaters
- Storage tanks
- Distribution pipe networks
- Recirculation pumps

Common applications: Hospitals, Hotels, Universities, High-rise residential buildings

Advantages:

- Efficient energy use
- Centralized maintenance
- Suitable for high demand

Disadvantages:

- Complex pipe networks
- Heat losses in long pipelines

b) Decentralized (Localized) Hot Water Systems

In decentralized systems, **multiple small heaters are installed close to points of use.**

Examples include: Electric shower heaters, Under-sink heaters

Advantages:

- Reduced distribution heat losses
- Lower installation cost for small buildings

Disadvantages:

- Multiple units require maintenance
- Higher long-term operational costs

Main Water Heating Systems

1. Centralized Hot Water System

In this system, **hot water is produced at a central location** and then distributed through pipes to various fixtures throughout the building.

Characteristics

- One main heater or boiler
- Large storage tanks
- Distribution network supplying the whole building
- Often includes **recirculation systems**

Typical Applications

- Hotels
- Hospitals
- Apartments
- Universities
- Industrial facilities

Advantages

- Efficient for large buildings
- Easier maintenance (central equipment)

Disadvantages

- Higher installation cost
- Heat loss in long pipe runs

2. Localized (Point-of-Use) Water Heating System

In this system, **water heaters are installed close to the fixture where hot water is required.**

Examples

- Electric shower heaters
- Under-sink heaters
- Small tank heaters in kitchens

Characteristics

- Small heaters installed at each location
- No large distribution pipes

Advantages

- Reduced heat loss
- Lower initial cost for small buildings
- Instant hot water at fixtures

Disadvantages

- Multiple units required
- Higher maintenance effort

3. Combined (Hybrid) Hot Water System

This system combines **central heating with localized heating** to meet varying demand.

Example

A building may have:

- **Central solar hot water system**
- **Electric point heaters for peak demand**

Advantages

- Flexible
- Energy efficient
- Reduces central system load

Applications

- Large apartments
- Hotels
- Hospitals
- Institutions

c) Hot Water Distribution Systems

Hot water distribution systems must maintain **adequate pressure, flow rate, and temperature** at all fixtures.

The main distribution methods include:

Gravity Circulation Systems

Hot water is supplied from **elevated storage tanks**.

The driving force is **gravitational head**:

These systems are simple but limited in tall buildings.

Pumped Circulation Systems

Circulation pumps maintain continuous water flow through the system.

Advantages include:

- Consistent pressure
- Suitable for tall buildings

Recirculating Hot Water Systems

Recirculation systems continuously circulate hot water in the distribution network.

This reduces **waiting time for hot water at fixtures**.

Benefits:

- Improves user comfort
- Reduces water wastage

However, it increases:

- Pumping energy consumption
- Heat losses

Heat Loss in Hot Water Systems

Heat losses occur through:

- Pipe conduction
- Tank heat losses
- Distribution inefficiencies

To reduce heat losses:

- Pipes should be **insulated**
- Distribution networks should be **short and efficient**

Temperature Control and Safety

Hot water systems must balance **comfort and safety**.

Typical temperature guidelines:

Application	Temperature
--------------------	--------------------

Domestic bathing	40–50°C
------------------	---------

Storage tanks	60°C
---------------	------

Hospitals	60–70°C
-----------	---------

Safety devices include:

- Temperature relief valves
- Pressure relief valves
- Thermostatic mixing valves
- Expansion tanks

Design Considerations for Engineers

These are mostly based on code recommendation such **National Building Code**.

Hot water system design requires evaluation of:

1. Peak Demand

Calculated based on:

- Number of occupants
- Type of building
- Usage patterns

2. Storage Capacity

Storage tanks must meet **peak demand periods**.

3. Pipe Sizing

Pipes must ensure:

- Adequate flow rates
- Minimal pressure losses

4. Energy Efficiency

Design strategies include:

- Solar heating integration
- Pipe insulation
- Efficient pumps

5. Maintenance Accessibility

Mechanical equipment should be placed in **accessible plant rooms**.

Challenges in Hot Water Supply Systems in Kenya

Some common engineering challenges include:

- High electricity costs
- Limited adoption of advanced technologies
- Poor maintenance practices
- Mineral scaling in heating elements due to hard water

- Inconsistent water supply in some regions

Solar water heating has emerged as a **major solution to energy cost challenges**.

Future Trends in Hot Water Systems

Emerging technologies include:

- Smart water heaters
- Heat pump systems
- Solar-thermal hybrid systems
- Energy recovery systems

These technologies aim to **reduce energy consumption and environmental impact**.

Water supply system design consideration:

A well-designed water supply system must provide a sufficient quantity of water at an acceptable pressure to all fittings, while ensuring the water remains potable (safe for consumption) and the system remains quiet and durable. Water supply system design involves the **engineering planning, sizing, and layout of components** to ensure **adequate quantity, pressure, quality, and reliability of potable water** in buildings.

In Kenya, design must align with:

- **National Construction Authority (NCA) guidelines**
- **National Building Code**
- **Kenya Bureau of Standards (KEBS) standards**
- Local county plumbing regulations (e.g., Nairobi City County)

The following are some of the design considerations;

- Water pressure
- Water flow rate
- Flow rate and pipe size
- System layout
- Connection to the mains supply
- Backflow
- Mains connection
- Pipe materials and specification

a) Water pressure

If water pressure is too low, this will be inconvenient for building users. If pressure is too high, this will lead to wastage of water, as well as high wear and tear on the system.

pressure is the force required to move water through the system and overcome friction.

- **Static Pressure:** The pressure when water is not flowing.
- **Residual (Flow) Pressure:** The pressure available at the fitting when water is flowing at the required rate.
- **Formula:** $P = \rho \cdot G \cdot h$
 - Where P = Pressure (Pascals), ρ = density of water (1000 kg/m³), g = gravity (9.81 m/s²), h = vertical height or "Head" (meters).
- In high-rise buildings (e.g., Upper Hill or Westlands), municipal pressure is often insufficient for top floors. Designers must use **Booster Pumps** or **Gravity Tanks** on mechanical floors.
 - *Minimum Pressure:* Typically checked on code for each fixture
 - *Maximum Pressure:* Should not exceed 5 bar prevent pipe bursts and "Water Hammer."

Typically, pressure limiting or pressure reducing valves will be used to control pressure in mains-supplied hot water systems or where high pressure may lead to problems such as burst pipes. Low pressure systems require few valves or controls. In low or unequal pressure systems, pressure can be increased to adequate levels by storing water in a header tank so that gravity can be used to create water pressure. Pressure can also be raised to adequate levels using a pressurizing pump, in which case it may be necessary to use pressure limiting and pressure reducing valves.

b) Water flow rate

As with water pressure, flow rates are crucial. A flow rate that is too high will result in water being wasted, whereas a flow rate that is too low will mean that sanitary fixtures and appliances don't work properly. The flow rate Q is the volume of water passing through a pipe per unit of time.

- **Loading Units (LU):** Since not all taps are used simultaneously, we use "Loading Units" to estimate **Design Flow Rate**. The LU are provided by code
 - *Example:* A washbasin = 1.5 LU; A Water Closet (WC) = 2 LU.
- **Formula (Continuity Equation):** $Q = A \cdot v$
 - Where Q = Flow rate (m³/s), A = Cross-sectional area of pipe (m²), v = Velocity of water (m/s).

$Q = \text{Volume} / \text{Time}$

Where:

- Q = flow rate (m³/s or L/s)

- **Design Velocity:** To prevent noise and erosion, water velocity should be kept between **0.75 m/s and 2.0 m/s**.

Flow rate is affected by:

- Water pressure
- Pipe diameters – The smaller the internal diameter of the pipe, the lower the pressure and flow rate.
- Water temperature – higher temperatures will tend to raise pressure and flow rates

c) System layout

In the design process, the layout of the plumbing system will largely follow room layout. Nonetheless, there are many things to consider which relate to Code compliance, building users' comfort, and sustainability. When planning a water supply layout, the following must be considered:

- **Pipe runs and lengths** – Keep pipe runs as short as possible. Pass pipes close to fixtures to minimize the number of branches and unnecessary elbows, tees and joints. Having longer pipe runs and more fixtures will reduce flow rate, increase heat losses, and increase use of materials
- **Point of entry into the building** – This should be into a utility space such as garage/laundry and include an accessible isolating valve, line strainer and pressure limiting valve (if required)
- **Water heating system** – Locate centrally to reduce the length of pipe runs to fixtures because longer pipe runs require more water to be drawn off before hot water is discharged. Install a separate point-of-use water heater for fixtures that are more than 10 m from the main water heater
- **Noise prevention** – Avoid running pipes over or near bedrooms and living areas.

d) Backflow

Backflow is the unplanned reversal of flow of water (or water and contaminants) into the water supply system. The system must be designed and used to prevent contamination from backflow.

e) Mains connection

Where the water source is a mains supply, the network utility operator is responsible for the water supplied to the property boundary. The property owner is then responsible for providing the pipework to bring the water into the building. An isolating valve must be fitted

at the point of connection to allow for maintenance and repair of the building's water supply system if required.

f) Pipe materials and specifications

The pipes used in a building must not contaminate potable water supply, and must be suitable for the water pressure, flow rate and temperature of water they will be carrying. This will be influenced by the materials used and also by other factors such as the wall thickness. Other considerations are durability, ease of installation, cost, and sustainability.

Pipe materials and components must not contaminate potable water. They must also be:

- suitable for the expected temperatures and pressures
- compatible with the water supply, to minimize the potential for electrolytic corrosion
- suitable for the ground conditions (if used underground) to minimize the potential for corrosion of the exterior of the pipe
- suitable for the local climate (if used outdoors) such as freezing conditions or atmospheric salt

So, when selecting of materials for water supply pipes, consider water pressure, water temperature, compatibility with water supply, durability, support, ease of installation, and cost.

Definition of pipe:

A pipe is a tube or hollow cylinder made of metal, plastic or any other materials used for conveyance of water, oil, gas and any other fluid substances from point to point.

Water pipes are tubes that carry pressurized and treated fresh water to buildings and as well as inside the buildings.

They are mostly made of polyvinyl chloride (PVC/UPVC), ductile iron, steel, cast iron, polypropylene, polyethylene, copper, or lead (the use of lead pipes however is no longer preferred due to the poisoning effect of lead as a hard metal.)

The pipes also come in different sizes and grades according to the intended use and the scale of operation they are being put to. However, there are other factors that also dictate the grade and material type to be used in a given area.

Development of pipes

Pipes have developed since the early ages where wood was used as a pipe material in Britain and china to deliver water from distant places to homes. The development is also linked to

the smoking pipes used by many people around the globe. With time the wooden pipes were found inefficient and susceptible to damage and rotting, leading to the invention of lead pipes in the early ages. However, the lead pipes lead to a very great hazard since ingestion of lead ions has devastating effects to the functioning of animals and their genetic disorders in general. The pipes have therefore evolved, through the copper and zinc pipes to the later invention of ductile iron pipes and galvanized iron which have been in use for long up to date. With the development of plastic molding technology and the need to be economical in material use, invention of plastic pipes has led to the manufacture of a vast variety of plastic pipes e.g. UPVC, PPR and polyethylene to transmit both drinking and waste waters. A lot of improvement is still being undertaken on the transmission aspects to help reduce the costs and at the same time improve the safety and quality of transmission depending on the vast climatic and geological conditions around the world.

TYPES OF PIPES USED IN WATER TRANSMISSION

Pipes found in waterworks systems in Kenya are generally of the following materials;

1. Galvanized iron (GI)
2. Cast/ grey iron
3. Ductile iron
4. Steel
5. UPVC (un-plasticized polyvinylchloride)
6. PVC(polyvinyl chloride)
7. Polyethylene (PE)
8. GRP (glass reinforced plastic)
9. Pre-stressed concrete, cylinder or non-cylinder (PSC)
10. Reinforced concrete cylinder (RC)
11. HDPE and LDPE
12. PPR

Other materials include ***copper and lead*** which tend to be found in service pipes, plumbing, common connections and other small diameter mains

FACTORS INFLUENCING THE CHOICE OF PIPE MATERIAL

1. Cost of material
2. Ease of handling and suitability of use
3. Chemical resistivity

4. Corrosion effect to the pipe
5. Frictional resistivity of the material to water flow
6. Strength of material
7. Ability to withstand high temperatures
8. Flexibility of pipe with soil type of a given area
9. Degradation when exposed to given environmental and physical conditions
10. Ability to withstand pressure from the supply source

CONTRAST BETWEEN DIFFERENT PIPES AND ADAPTATIONS TO THE APPLICATION AREAS

- a. GI – Galvanized iron
- b. PPR – polypropylene Random pipes
- c. HPDE – High density polypropylene
- d. UPVC – unplasticized polyvinyl chloride
- e. PVC – polyvinyl chloride
- f. PE - polyethylene
- g. ABS – acrylonitrile butadiene styrene
- h. CPVC – post chlorinated polyvinyl chloride
- i. PB - polybutylene
- j. PP – poly propylene
- k. PVDF – polyvinylidene fluoride

GI PIPES

They are used mainly as the supply pipes on the main water lines for treated .mostly medium pipes are used. They vary in diameter from 15mm to 150mm. LENGH MOSTLY 6METRES



CASES WHERE THEY ARE MOSTLY USED

1. Rocky grounds.
2. Places where the pipeline crosses roads and busy paths.
3. Where water theft needs to be controlled

Advantages

- i. They are robust
- ii. Are more difficult to tap into illegally by those who want to steal
- iii. Are easy to join

Disadvantage

- i. Are so expensive
- ii. They are very disruptive and difficult to lay
- iii. Corroded in some soils
- iv. Unsuitable for drinking water when the metal gets corroded.

DUCTILE IRON PIPES

Also used as distribution pipes. Similar to galvanized iron, they are robust and difficult to tap into illegally.

Disadvantages

- i. Need to coated depending on the soil to prevent rusting

- ii. Rubber gaskets at joints can be damaged as pipes are joined leading to contamination
- iii. Are expensive

ALLUMINIUM PIPES

Aluminum tubes used to be more popular on the market as a pipe due to its light weight, durable and easy construction, which can be more suitable for using in home improvement. It gradually did not have the market. The products are however being phased out gradually.

Advantages:

- i. The price is cheaper.
- ii. Can be arbitrary curved bow.
- iii. Smooth surface.
- iv. Construction is convenient.

Disadvantages:

- i. Easy to aging
- ii. The pipe joint leakage phenomenon appears easily.
- iii. Have a very short life span

PPR PIPES. (POLYPROPYLENE RANDOM PIPES)

PPR pipes are designed for Hot and Cold water supply and heating applications and it is suitable for different applications listed below:

- i. Hot and cold water supply in residential,
- ii. industrial, commercial & public projects
- iii. Solar applications
- iv. Compressed air systems
- v. Drinking water and liquids
- vi. Watering systems for greenhouses & gardens
- vii. Transportation of aggressive fluids
- viii. Water purifying plants
- ix. Radiator heating
- x. Traditional heating systems

Their sizes range from 20mm to 110mm in thicknesses while lengths vary too e.g. mostly 4 meters and in rolls of 100 meters.



FEATURES AND BENEFITS

1. Light weight, easy and quick assembly
2. Most suitable for carrying drinking water
3. Excellent corrosion and chemical resistance
4. Low thermal conductivity
5. Safe and watertight joints
6. Reduce heat loss
7. High impact strength
8. Resistance to scaling
9. Resistance to frost
10. Usable in seismic areas
11. Resistance to abrasion
12. Resistance to stray current
13. Eco-friendly
14. Long operational durability
15. Economical in the overall capacity.

They are safe and reliable can be used up to 50 years.

Disadvantages:

- I. The construction technical requirements higher.
- II. They need to use special tools and professionals to undertake construction in order to ensure system safety. Bacteriologically neutral

HDPE Pipes (High density polythene)

High-density polyethylene (HDPE) is a polyethylene thermoplastic made from petroleum. It is known for its large strength-to-density ratio. The density of HDPE can range from 0.93 to 0.97

g/cm³ or 970Kg/m³. The difference in strength exceeds the difference in density, giving HDPE a

higher specific strength. It is also harder and more opaque and can withstand much higher temperatures (120 °C for short periods, 110 °C continuously). High-density polyethylene, unlike

polypropylene, cannot withstand normally required autoclaving conditions. The lack of branching is ensured by an appropriate choice of catalyst (e.g., Ziegler-Natta catalysts) and reaction conditions.

HDPE pipes are best preferred for industrial, Domestic and irrigation purpose.

They have been preferred over other available resources due to varied salient features:

1. Strong and resilient
2. Light weight
3. Length as required
4. Better flow characteristics
5. Hygienic & Odourless
6. Energy saving
7. Leak proof
8. Chemical Resistant
9. Economical
10. Long Lasting
11. Maintenance free

They are effectively applied in water supply in irrigation field, sprinkler irrigation, bio-gas transportation, drawing water from pump set for distribution, most suitable for submersible pump

and jet pumps and distribution of water in water projects. Their sizes vary from about 16mm to 110mm.

Disadvantages

- I. Can't be used in certain conditions of soil e.g. that contaminated with oils and petrol
- II. Need to be joined together correctly to avoid contamination



UPVC – UNPLASTICISED POLYVINYLCHLORIDE

UPVC pipes with Elastomeric Sealing Ring type joints are best preferred for supply of water for rural and urban areas, for irrigation and water supply in hilly areas where temperature is very low and in desert areas where the temperature is at maximum.

Elastomeric Sealing Ring fit pipes have been preferred over other available resources in the mentioned areas due to the following salient features.

1. Strong and durable
2. Light weight
3. Convenient joining
4. Better flow characteristic
5. Energy saving
6. Leak proof
7. Resistant to rusting, chemical action, weathering & scale formation
8. Odourless and hygienic
9. Long lasting & maintenance free



APPLICATIONS

Domestic: Supply of potable water in house and residential buildings.

Agriculture: Supply of water for irrigation of crops.

Other: Supply of water for irrigation and consumption in hilly areas where temperature is very low and in desert areas where the temperatures are the maximum

Disadvantages

- I. Can be damaged by exposure to sunlight
- II. Joints are glued making them prone to leakages
- III. Need proper bedding when laying e.g. sand
- IV. Pressure from sharp objects can puncture it leading to leakages and contamination

PVC PIPES

PVC pipe has made a tremendous improvement in plumbing. It replaces cast iron and galvanized pipe in almost all situations. Light weight and easy to work with, PVC is available in many different sizes. Fittings and related materials are readily available at all plumbing and hardware stores. Furthermore, they are more affordable for our economy and hence their preference over metal pipes.



What is PVC?

PVC pipe is made from the plastic, “polyvinyl chloride.” It is used in construction of drains, as

vents, and to handle waste in buildings. It is rigid, lightweight, and strong. Because of PVC pipe's ease of installation, it is ideal for drain applications under kitchen sinks and bathroom vanities. The many fittings available for attaching PVC make it universal in all settings except

very high temperature applications. It works much better for plumbing than the old standard cast

iron pipe because it does not need to be hot soldered, is resistant to almost any alkaline or toxic

substance, and is easy to install. There are two types, which are defined in standards -- type 40,

for personal homes, and type 80, used in industrial settings.

Some Advantages

- Easy to install and hence low installation cost
- chemically resistant
- Strong
- Fire resistant
- High internal corrosion resistance
- Immune to galvanic or electrolytic attack
- Free from toxicity, odorless and tasteless
- Low friction loss

- Low thermal conductivity

Various matters of concern about PVC are; design life of pipe, possibility of recycling, resistance to UV rays, resistance to fire etc.

Design life-PVC, because of its composition, will last for the design life of the home in which it

is installed. Because it is rustproof and chemical resistant, a design life of 100 years is typical.

Recycling - The pipe can be pulverized and returned to the extrusion process to manufacture new

pipe. There are currently no standards for this. Because of its long life, PVC hasn't had a lot of

exposure to recycling.

Resistance to UV rays - Because of its inherent design PVC pipe contains stabilizers that protect

the pipe against attack by UV rays present in sunlight. Some discoloration may occur. Several years of exposure may see slight reduction in the impact resistance of the material. By

painting

the exposed pipe with a latex paint (don't use oil based) this problem is virtually eliminated.

Fire resistance - Like any plastic, PVC pipe will melt if subjected to high temperatures.

However, it stops that process immediately when the fire source is removed. Studies show that

PVC pipe in a typical installation is less than 1 percent of all combustible materials in a building

Comparison between PVC and UPVC pipes

The main difference between PVC and UPVC is that UPVC doesn't contain phthalates or BPA.

This makes it safer for transporting water, as well as making it fire-resistant. Because of the concerns of plasticizers being ingested by people and animals it is better to ensure that only

UPVC pipe is used in potable water applications. For the purpose of drain pipe it is

acceptable to use straight PVC pipe. PVC pipe is often used to distribute water that people aren't going to drink. It's also used to insulate electric cables. The UPVC versions often

replace wood when building window frames and sills. The UPVC pipe variant also often

replaces pipes made of cast iron for drains, waste piping, downspouts and gutters.

Many people use PVC instead of metal, as PVC is much easier to cut than metal, and it's also

easy to use glue together. In addition, PVC is still fairly tough, so not much strength is lost. Most plastic piping around the globe is actually UPVC because of how resistant it is to degradation caused by chemicals, high and low temperatures and various pressure points. The UPVC version of piping is less flexible than regular PVC, but it's also more recyclable.

Application

UPVC is used in transmission of drinking water while PVC is best suited for waste water transmission from various points of use

PE PIPES

Have the following unique characteristics

1. High density
2. Good flexibility
3. Bear strong shocks and twists in earthquakes
4. Long performance life of 50 year
5. High density, good flexibility, bears strong shocks and twists in earthquakes, resists wear and tear
6. Shock resistant
7. Does not cause bacteria
8. Will not cause secondary pollution
9. Has good corrosion resistance
10. Resists fractures and handles extreme temperatures (-40 ~ 40oC)
11. Has long performance life and can be used for 50 years under normal situations
12. Lightweight and easy to weld
13. Safe connections
14. Clean and non-toxic

USES

- a) Mainly used for municipal administration water supplies, buildings' water
- b) supply, chemicals, food, material, prints, pharmaceutical, light industry,
- c) papermaking and metallurgy
- d) For water and other liquids
- e) Used in irrigation, telecommunications lines, sleeve pipes, as well as all facets
- f) of chemical industry
- g) Liquid transports pipeline of industry and irrigates pipeline for farming

Disadvantages

- i. Can be damaged by exposure to sunlight
- ii. Joints are glued making them prone to leakages
- iii. Need proper bedding when laying e.g. sand
- iv. Pressure from sharp objects can puncture it leading to leakages and contamination

INSTALATION

Pipe installation one of the most important aspects that determine the *durability of the pipelines and hence the conservation of water and prevention from contamination.*

Precaution therefore has to be taken so that this is done in the most perfect way to conserve the pipes for a longer duration. A proper analysis is done in chapter 4 of this project.

Procedure

1) PVC can be cut easily. It can be cut with a hacksaw, but abrasive disks are made for miter saws that work better to get a straight edge. A joint that is skewed due to pipe not being cut straight can throw off the entire run of pipe.

2) After cutting, all shavings are cleaned out of the pipe and the inside edges deburred. When the pipe is cut to the proper length, it's laid out on the floor with fittings in place to determine if the length is correct. If the length is found proper, installation can proceed.

3) The pipe must be cleaned with all-purpose pipe cleaner, called primer. The primer is swabbed around the end of the pipe and the inside of the fitting to ensure there are no contaminants that can get in the way of adhesion. PVC is joined with a special type of cement. The cement sets up very quickly, so you must be ready to go as soon as it is applied. The inner surface of the joint is coated with cement then the pipe inserted and turned round to ensure the glue has covered the entire joint. Care should be taken to ensure the pipe is seated correctly in the joint.

4) Once the PVC pipe is in place, and proper length has been determined, pipe hangers are installed to support the pipe. This eases strain on the joints that could lead to possible leakage. Recommended distances from hanger to hanger, usually every 4 feet is used, allowing for movement in expansion and contraction. The pipe must be protected from nails & sharp objects.

PIPE FAILURE

They are mostly linked to improper installation practices and methods in the field.

Main Causes of Pipe Failure are:

1) Improper System Engineering/Installation

- i. Inadequate provision for linear thermal expansion
- ii. Excess use of Cement
- iii. Insufficient amount of Cement
- iv. Wrong Clamps used or Clamps too tight
- v. Incompatible fire caulk used
- vi. Contact of outside of pipe with incompatible material (e.g., solder flux)

2) Improper Operation

- i. Exposure to freezing temperatures without freeze protection
- ii. Over-pressurization
- iii. Pulsating water pressure
- iv. Use of incompatible materials around pipes

3) Contamination

Both Internal and external for example,

- i. Use of contaminated antifreeze
- ii. Contaminants from metal water supply piping; e.g., antimicrobial (MIC inhibitor) linings.
- iii. Incompatible Fire Caulk
- iv. Use of incompatible (black Proset) grommets to seal pipe against hole in concrete.
- v. Contact with incompatible plastic coated wires
- vi. Exposure to hot solder flux
- vii. Exposure to hot polyurethane foam insulation

4) Manufacturing defects

- i. Dirty extrusion die
- ii. Incomplete resin consolidation
- iii. High stresses in pipe wall due to rapid cooling
- iv. Occlusions, char particles, voids
- v. Filler/pigment not well distributed

5) Abuse by Distributor

- i. Store in sun
- ii. Damage during transport due to careless handling.

PROBLEMS ASSOCIATED WITH PIPELINE TRANSMISSION OF WATER.

- i. Bursting of pipes leading to water wastage.
- ii. Friction generation as water moves in the pipes.
- iii. Dynamics with pressure changes with the altitude.
- iv. Stealing of the pipes by the public.

- v. Effects of corrosion e.g. rusting.
- vi. Danger of contamination in case of a burst.
- vii. Theft of water pipes as well as illegal connections.
- viii. The difficulty of forecasting the future demand of water.
- ix. Contamination of water through some points.
- x. Insufficient water.
- xi. Freezing of water in pipes

PRICES

The GI pipes are very expensive when compared to the plastic ones. The HDPE pipes are sold in terms of cash per unit weight but project virtually higher than the other plastic pipes followed by the PPR and the PE pipes.

UPVC is the lowest in price of the plastic pipes and it's also found in much larger metric series of up to over 20 inch.

Major types of valves used in water supply pipework

A valve is a type of fitting that allows for regulation, control, and direction of fluids passing through a pipe. Valves are commonly used to direct flow, start or shut off water access, prevent backflow, isolate sections of piping for maintenance and adjust/regulate water pressure within a system. Valves allow homeowners to isolate sections of plumbing for repairs, or shut off a water line when a leak sprouts.

Main Types of Plumbing Valves

The most common valves used in plumbing systems are:

1. Stop valve
2. Gate valve
3. Globe valve
4. Check valve (Non-return valve)
5. Ball valve
6. Float valve
7. Pressure reducing valve
8. Foot valve
9. Butterfly valve
10. Air valve

1. Stop Valve (Stopcock)

A stop valve is a valve used to completely stop or allow the flow of water in a pipe.

They are commonly installed: at the property boundary, point where the supply pipe enters the building, before plumbing fixtures

The stop valve is used to:

- Shut off water supply during repairs
- Isolate sections of plumbing
- Control water supply to buildings or fixtures

Mostly found Inside the meter box and the main entry point into the house

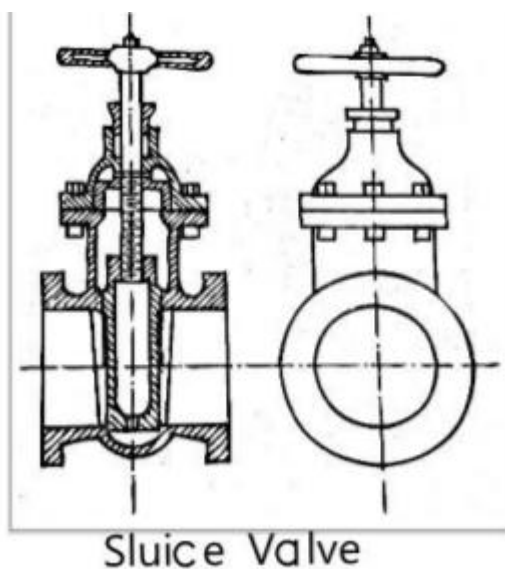
2. Gate Valve/sluiice valve

Sluice valves are also known as gate valves, cut off valves or shut off valves. They are required to shut off the supplies whenever desired in the water supply system.

It uses a **metal gate or wedge** that moves up and down to start or stop the flow of water.

Mechanism: The valve consists of a gate connected to the wheel by the spindle. When the wheel is rotated in an anti-clockwise direction the spindle rises up and allows the water to flow through it. When the wheel is rotated in clockwise direction, the flow of water is cut off.

Gate valves cannot control the water flow as they are designed to be fully opened or fully closed. If used to adjust the water flow, it can wear out the valves. Gate valves can be used for the circulation of all types of fluids, such as air, fuel gas, feedwater, steam, lube oil, and hydrocarbons.



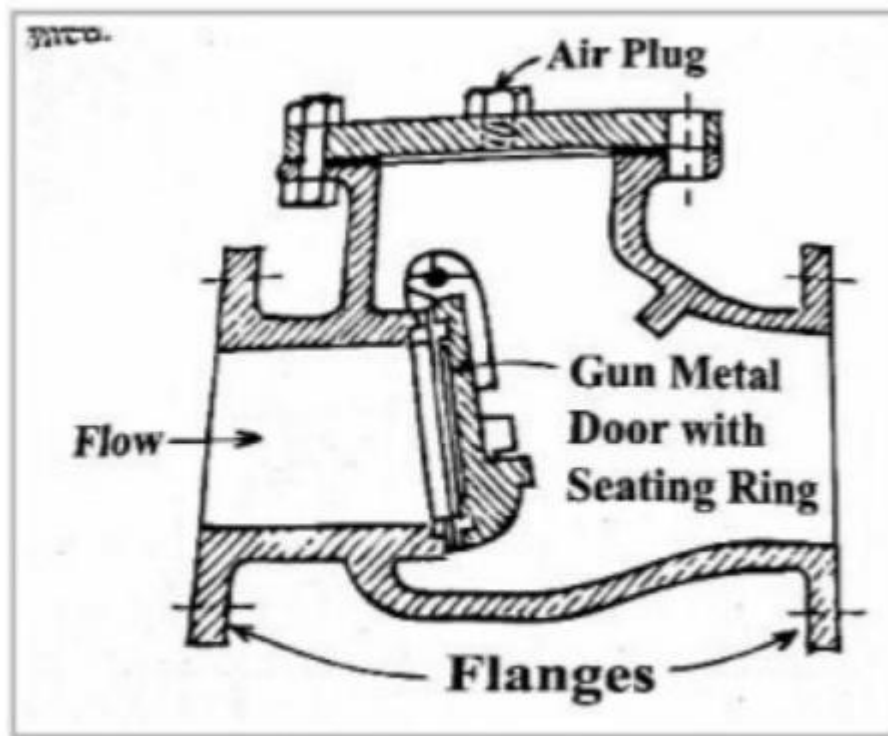
3. Reflux/ Check Valve (Non-Return Valve)

A check valve is a valve that allows water to flow in only one direction.

It automatically prevents reverse flow.

The check valve prevents; Backflow contamination, Reverse flow in pumps, Drainage of water from tanks. Example: Installed after a water pump to prevent water from flowing back.

Mechanism: It consists of a hinged gate. When water flows the pressure of flowing water make the gate open by rotating about the hinge and allows the water to pass through the other section. When the flow of water stops the gate gets closed by rotating about the hinge and prevents the backflow of water.



Reflux Valve

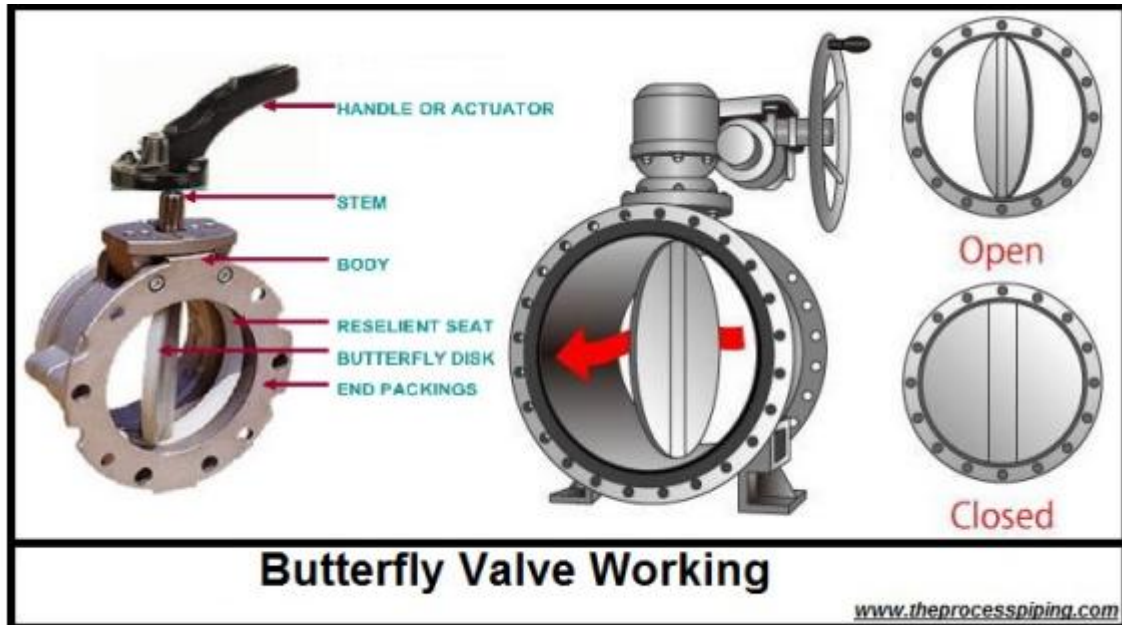
4. Butterfly Valve

This valve has a rotating metal disc that allows and inhibits the water flow, creating an image similar to that of a butterfly due to which is called a butterfly valve.

These valves are very compact, light, and relatively short, making them significantly lighter than the other types. The main components of a butterfly valve are the body, disc, stem and seat.

Mechanism: The valve is fully open or closed when the disc is rotated a quarter turn. The “butterfly” is a metal disc mounted on a rod. The disc is positioned in the center of the pipe. A rod passes through the disc to the outside of the valve. Rotating the rod, through hand-wheel or actuator, turns the disc either parallel or perpendicular to the flow. When the disc

plane is at right angle or perpendicular to the center line of pipe, butterfly valve is closed. Then when the disc is rotated a quarter turn to make the plane of disc in line with the center line of pipe, butterfly valve is fully open and it allows an almost unrestricted passage of the fluid. The valve may also be opened incrementally to throttle flow.



5. Ball Valve

Ball valves are the most reliable and common type of valves used to regulate the flow of water. It involves a rotating sphere with a hole that is attached to a lever handle to operate the valve. In the open position, the hole in the sphere is in line with the pipe, allowing the water to flow through it. When in a closed position, the hole in the sphere is perpendicular to the pipe, thus completely restricting the water flow. The lever handle also serves as an indicator of whether the valve is open or closed. When the lever is perpendicular to the pipe, the valve is closed.

Ball valves are commonly used in: Domestic plumbing systems, Water tanks,

Float Valve

A float valve is an automatic valve used to control the water level in storage tanks.

It operates using a floating ball connected to a lever. When the water level drops, the valve opens and when the tank fills, the float rises and closes the valve.

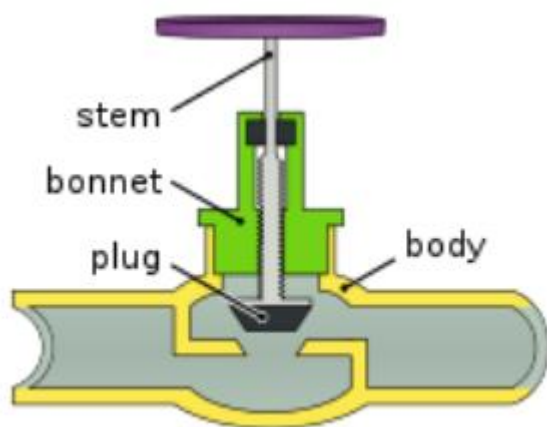
Float valves are commonly installed in: Overhead tanks, underground water tanks, toilet cisterns.

Globe valve

The Globe valve is commonly used to regulate or limit the water flow in plumbing applications, where the flow needs to be adjusted regularly. Unlike the gate valve, which cannot be used for this function

The interior design features contain a stopper on the end of a valve stem that is raised and lowered by the valve's twist knob. Globe valves get their name due to the globe-like or ball-like appearance of their body.

The design of the globe valve body forces the flow of the commodity to change direction within the valve itself. This change in direction creates substantial pressure drop and turbulence. The globe valve is therefore not recommended when flow resistance and pressure drop are to be avoided.



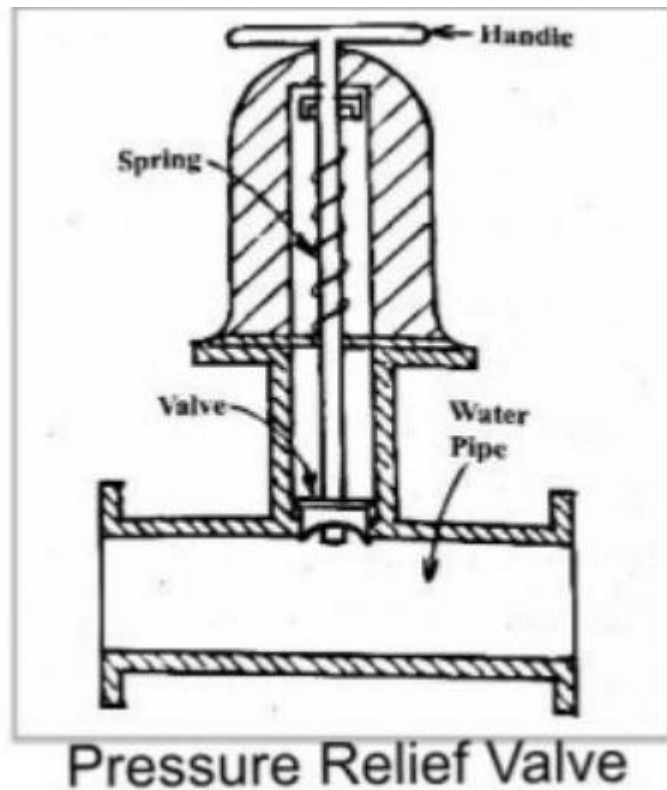
Pressure Relief Valve

Pressure relief valves are also known as automatic cutoff valves or safety valves. They are used to keep the pressure below predetermines value in the pipeline. When the pressure exceeds the predetermine value there exist the chance of bursting of the pipes, so pressure relief valves are used.

High water pressure can: damage pipes, cause leaks and damage plumbing fixtures

PRVs are commonly installed at building entry points, in high-rise buildings and in areas with high municipal pressure

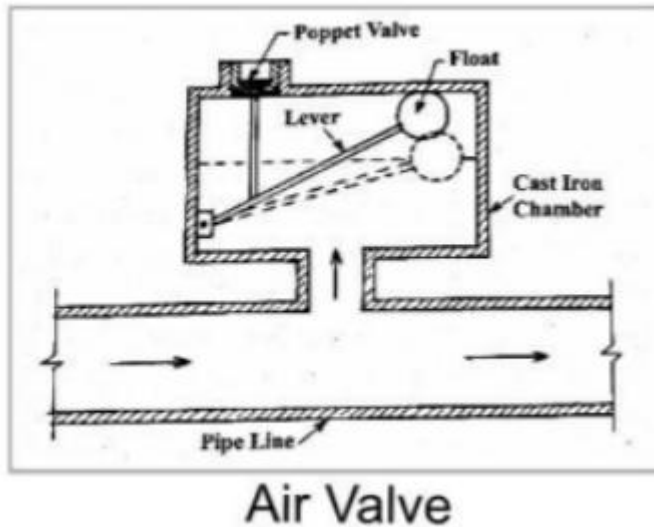
Mechanism: The valve consists of the spring-loaded disc which is adjusted to a predetermined pressure. When the pressure in the pipeline is below the predetermined value, the valve remains closed. But when the pressure exceeds the predetermined value, the valve opens as the disc is lifted up and the certain amount of water is discharged which reduces the pressure.



Air Valves

Air valves are provided to prevent the accumulation of air at higher points that creates backward pressure. So, air valves are provided at higher points to make the accumulated air exit.

Mechanism: It consists of the cast iron chamber, float, lever, poppet valve. When there is no air present in the iron chamber, the float is connected to the roof. When the air start accumulating in the chamber the float gets lowered and the poppet valve gets opened and the air gets escaped. The float again moves to the original position (touches the roof) after the air is liberated.



Fittings used in water supply pipework

Fittings are appurtenances used in the pipeline to carry out the following functions:

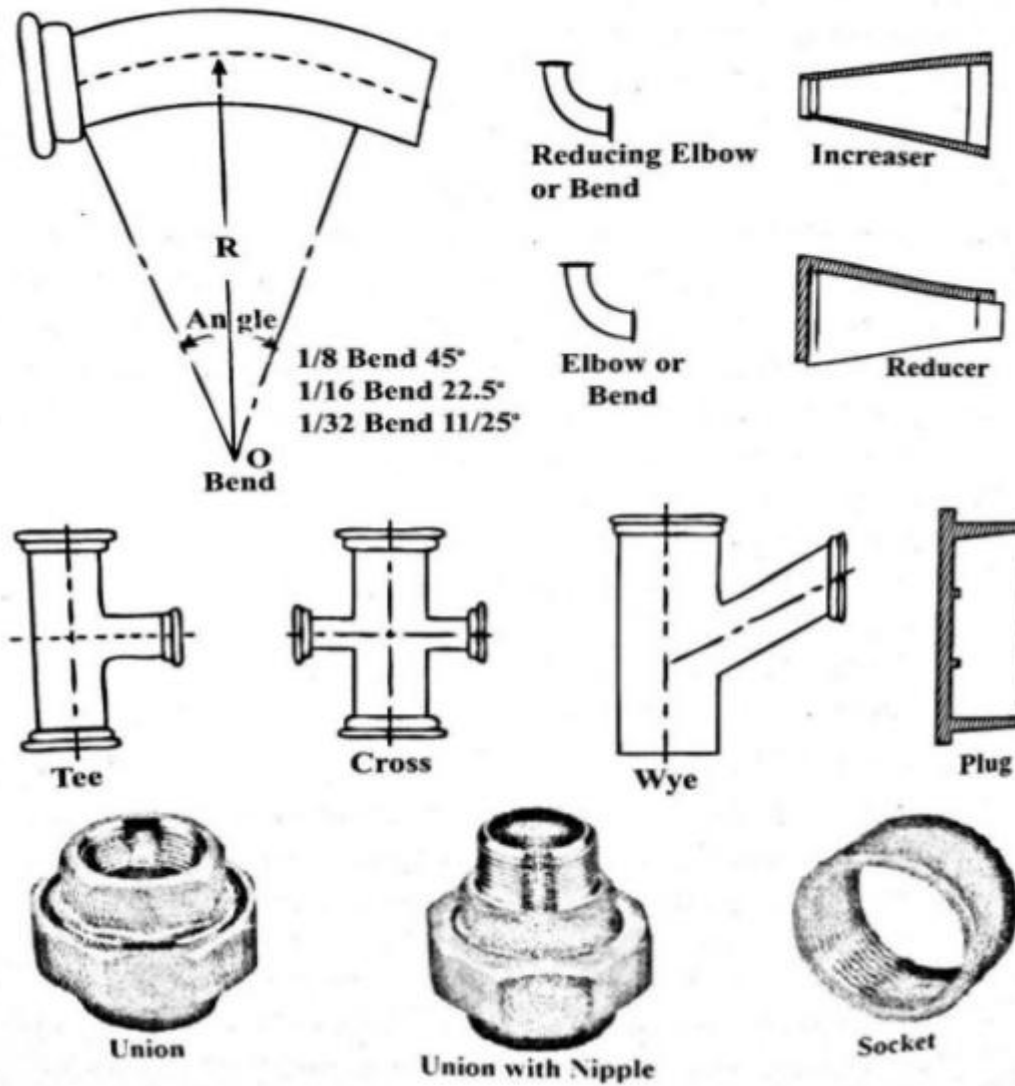
- To connect pipe of different sizes
- To connect pipe to other appurtenances
- To change the direction of flow
- To change the gradient of flow
- To connection pipe sections

- 1) **Bend:** Bend is a fitting used to change the direction of a pipeline.
- 2) **Tee:** Tee is a fitting with one inlet and two outlets. It is used to for connecting pipes of different diameters or for changing the direction of pipelines.
- 3) **Cross:** Cross is a fitting to connect four pipe sections. It has one outlet and three inlets or vice versa.
- 4) **Wye:** Wye is a fitting with two inlets or one outlet or vice versa. It is used to create a branch.
- 5) **Reducer:** Reducer is a fitting that joins two pipes of different diameters.
- 6) **Plug:** A plug is a fitting that closes off the end of the pipe.
- 7) **Socket:** Socket is a fitting that fits over the pipe. It is classified as the plain and reducing socket. Plain socket connects two pipes of same diameters. Reducing sockets is used to connect two pipes of different diameters.
- 8) **Nipple:** Nipple is a fitting (short stub of pipe) which is used for connecting two other fittings.
- 9) **Union:** Union is a fitting to connect two pipes of the same diameter for quick and

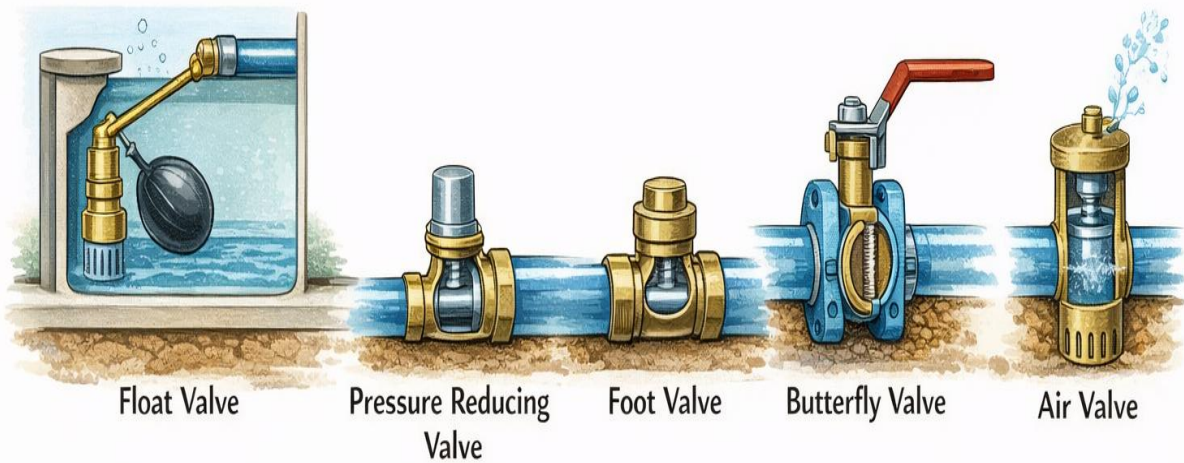
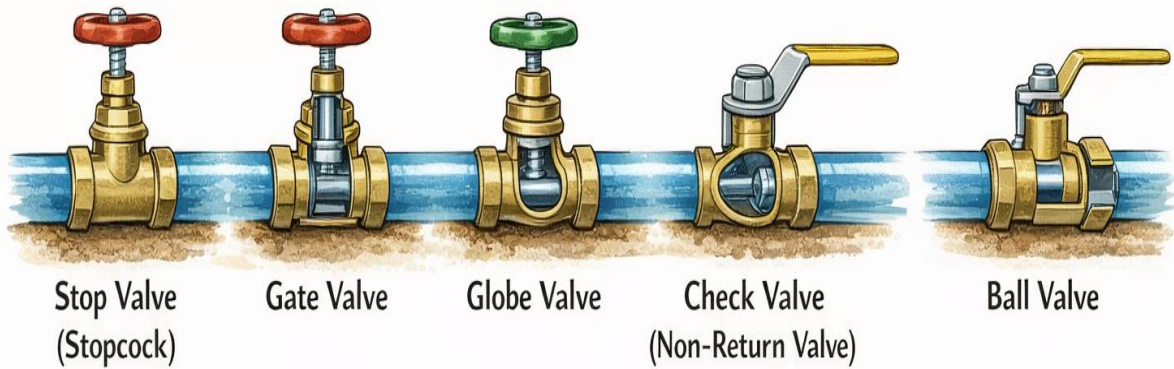
convenient disconnection.

10) **Stopcock:** Stopcocks are fittings which are cut off valves of small size.

11) **Water tap:** Water taps are fittings which are used to obtain water from the consumers.



Types of Valves Used in Plumbing



Stop Valve
(Stopcock)

Pressure Reducing Valve

Foot Valve

Air Valve